Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 Chromium is obtained from the following reaction, $2Al + Cr_2O_3 \longrightarrow 2Cr + Al_2O_3$. What is the percentage yield of chromium when 180 g of chromium is obtained from a reaction between 100 g of aluminium and 400 g of chromium(III) oxide?

A 45.0%

B 36.0%

C 93.6%

D 131%

2 0.5 g of zinc powder was found to reduce an acidified solution of 25.50 cm^3 of 0.200 mol dm^{-3} $\mathbf{A}O_2^+$. What is the final oxidation number of metal **A**?

A 0

B +1

C +2

D +3

3 In an experiment, a sample of Americium element (Am) was vaporised, ionised and passed through an electric field. Analysis revealed that a beam of $^{241}Am^+$ gives an angle of deflection of 2^o. What would be the angle of deflection for a sample of singly charged calcium ions?

A 12.0^o

B 6.0^o

C 2.0^o

D 0.3^o

4 The use of the *Data Booklet* is relevant to this question.
E and **G** are in Group V and Group VI respectively. Which of the following comparisons between the first and second ionisation energies of **E** and **G** is correct?

	1^{st} I.E.	2^{nd} I.E.
A	**E** < **G**	**E** > **G**
B	**E** < **G**	**E** < **G**
C	**E** > **G**	**E** > **G**
D	**E** > **G**	**E** < **G**

5 The successive ionisation energies (I.E.) of two elements, **J** and **L**, are given below.

I.E./ $kJ\ mol^{-1}$	1^{st}	2^{nd}	3^{rd}	4^{th}	5^{th}	6^{th}	7^{th}	8^{th}
J	1090	2350	4610	6220	37800	47000	-	-
L	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **J** reacts with **L**?

A **JL**

B $\mathbf{J_2L_3}$

C $\mathbf{JL_4}$

D $\mathbf{J_4L}$

6 Three substances **Q**, **R** and **T** have physical properties as shown.

Substance	Melting point/°C	Boiling point/°C	Electrical Conductivity	
			of solid	of liquid
Q	801	1413	Poor	Good
R	2852	3600	Poor	Good
T	3550	4827	Good	Unknown

What could be the identities of **Q**, **R** and **T**?

	Q	**R**	**T**
A	$AlCl_3$	NaC*l*	C(diamond)
B	$AlCl_3$	MgO	C(graphite)
C	NaC*l*	MgO	C(graphite)
D	NaC*l*	MgO	C(diamond)

7 Which one of the following pairs contain substances with similar shape?

A CO_2 and Cl_2

B NH_3 and NO_3^-

C $BeCl_2$ and $SnCl_2$

D BCl_3 and PCl_3

8 The reaction between **U** and **X** is as follows:

$$U + 2X \longrightarrow 2Y$$

In an experiment to investigate the effect on the concentrations on rate of reaction, the following results were obtained at constant temperature.

Experiment	[**U**]/ mol dm^{-3}	[**X**]/ mol dm^{-3}	Initial rate/ mol dm^{-3}s^{-1}
1	1.0	1.0	0.0008
2	1.0	2.0	0.0016
3	1.0	3.0	**Z**
4	2.0	2.0	0.0032

What is the value of **Z**?

A 0.0008

B 0.0016

C 0.0024

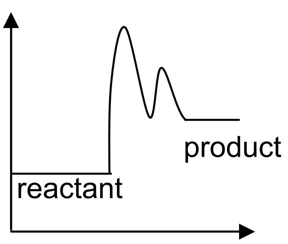
D 0.0032

9 The following reaction proceeds by two stages:

$$\text{Reactants} \xrightarrow[\text{(slow)}]{\text{step I}} \text{Intermediate} \xrightarrow[\text{(fast)}]{\text{step II}} \text{Products} \qquad \Delta H > 0 \text{ kJ mol}^{-1}$$

Which one of the following reaction pathway diagram correctly represents the energy changes in the above reaction?

A



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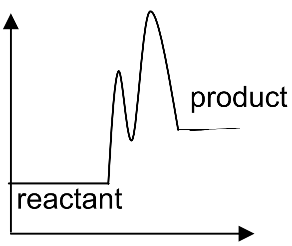

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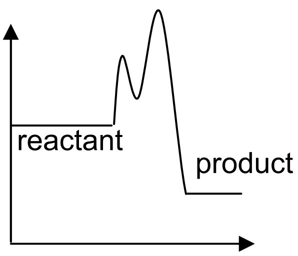

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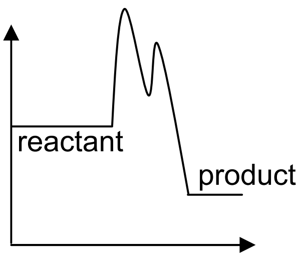


10 Which of the following statements is always **true**?

A At a given temperature, all molecules possess the same kinetic energy.

B Halving the pressure of a gaseous reaction increases the rate of reaction.

C The rate of a reaction increases as the reaction proceeds.

D A catalyst increases the rates of both forward and backward reactions for a reversible reaction.

11 When chlorine gas dissolves in water, a solution smelling of Cl_2 is produced:

$$Cl_2(g) + H_2O(l) \rightleftharpoons HClO(aq) + HCl(aq)$$

Which of the following will remove the smell of Cl_2?

A addition of concentrated HC*l* to the solution

B addition of NaOH to the solution

C decreasing the pressure of the system

D adding NaC*l* and allowing it to react with water

12 Which of the following statements regarding enthalpy change is **true**?

A ΔH_{rxn}^{θ} is the same regardless of the reaction pathway taken as long as the initial and final conditions are the same.

B ΔH_{c}^{θ} is the energy change when one mole of the substance is burnt in air at 1 atm and 298 K.

C ΔH_{n}^{θ} is the amount of energy released when 1 mole of acid reacts with 1 mole of base at 1 atm and 298 K.

D ΔH_{f}^{θ} is the energy change when 1 mole of substance is formed from its constituent elements at 1 atm and 293 K.

13 The standard enthalpy change of the following reaction is –896.4 kJ mol^{-1}.

$$2HN_3(l) + 2NO(g) \longrightarrow H_2O_2(l) + 4N_2(g)$$

	NO(g)	$H_2O_2(l)$
ΔH^{θ}_{f}/kJ mol^{-1}	+90.3	–187.8

Using the given information in the above table, what is the standard enthalpy change of formation, in kJ mol^{-1}, of $HN_3(l)$?

A +264

B +528

C +618

D +632

14 The table shows the enthalpy change of neutralisation per mole of water formed, ΔH, for various acids and bases.

Acid	Base	ΔH / kJ mol^{-1}
hydrochloric acid	sodium hydroxide	–57.0
E	sodium hydroxide	–54.0
hydrochloric acid	**F**	–52.0
nitric acid	**G**	–57.0

What are **E**, **F** and **G**?

	E	**F**	**G**
A	ethanoic acid	ammonia	potassium hydroxide
B	ethanoic acid	sodium hydroxide	ammonia
C	sulfuric acid	ammonia	potassium hydroxide
D	sulfuric acid	sodium hydroxide	ammonia

15 Given the following data, what is the standard enthalpy change of combustion of methanol, CH_3OH, in kJ mol^{-1}?

ΔH_c (graphite) = –394 kJ mol^{-1}
ΔH_f (water) = –286 kJ mol^{-1}
ΔH_f (methanol) = –239 kJ mol^{-1}

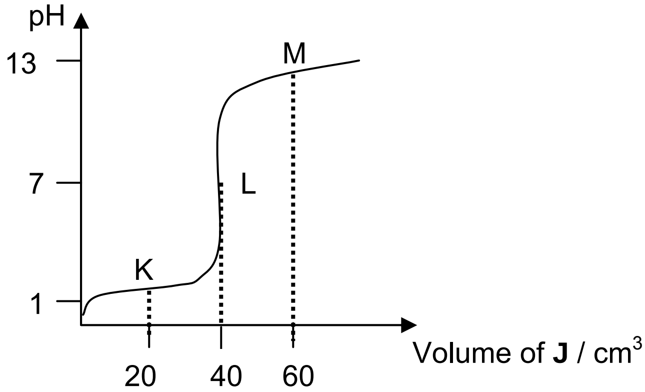
A –441
B –727
C –919
D –1025

16 An enzyme, found in stomach, operates at maximum efficiency when in an aqueous solution buffered at pH 5.

Which combination of substances when dissolved in 10 dm^3 of water, would give the necessary buffer solution?

A 1 mol HC*l* and 1 mol of CH_3COOH
B 1 mol CH_3COOH and 1 mol of CH_3COONa
C 1 mol HC*l* and 1 mol of CH_3COONa
D 1 mol of CH_3COONH_4 and 1 mole of NH_3

17 The following graph shows the pH changes when a 0.10 mol dm^{-3} solution of **J** is added to 20.0 cm^3 of 0.10 mol dm^{-3} of sulfuric acid.



Which of the following statements regarding the titration is **correct**?

A It is a weak monoprotic base.

B At point L, the salt at end point is hydrolysed to give an acidic solution.

C Phenolphthalein is a suitable indicator for this titration, but not methyl orange.

D There are no buffer regions in this graph.

18 The value of the ionic product of water, K_w, varies with temperature.

Temperature / °C	K_w / mol^2 dm^{-6}
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

A The ionic dissociation of water is an exothermic process.

B At 62 °C, water with a pH of 6.5 is neutral.

C The pH of pure water increases with temperature.

D The association of water molecules by hydrogen bonding increases as temperature increases.

19 Consider the sequence of oxides Na_2O, SiO_2, P_4O_{10}.
Which factor decreases from Na_2O to SiO_2 and also from SiO_2 to P_4O_{10}?

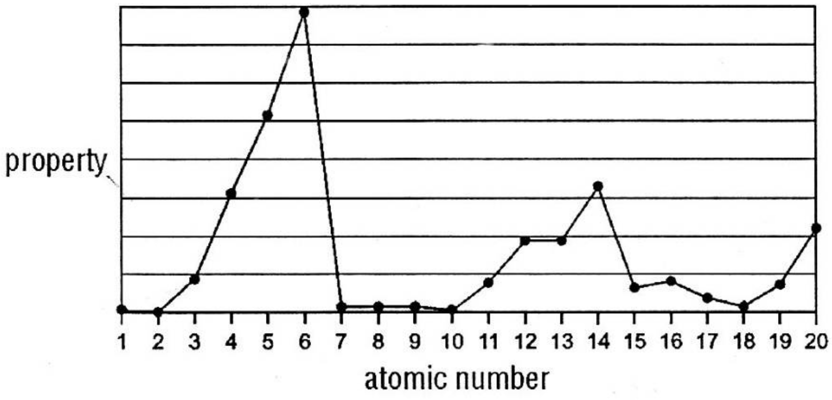
A covalent character

B electrical conductivity in the solid state

C pH when mixed with water

D solubility in aqueous alkali

20 The following shows the variation of a property of the first 20 elements in the Periodic Table with the atomic number of the element.



What is the property?

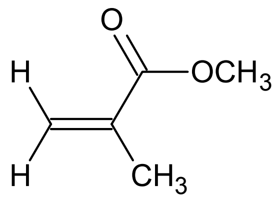
A Atomic radius

B First ionisation energy

C Ionic radius

D Melting point

21 Which of the following statements is **true** of the organic compound shown below?



A It is a cis isomer.

B It has three sp^2 carbon atoms and two sp^3 carbon atoms.

C It has 7σ and 2π bonds.

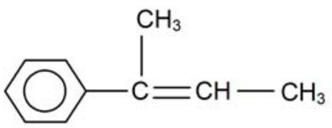
D It will form a ketone as one of the products with hot acidified $K_2Cr_2O_7$(aq).

22 Compound **E**

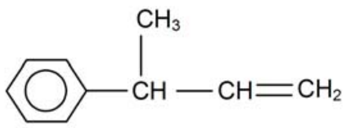
- has the molecular formula $C_{10}H_{14}O$;
- is unreactive towards mild oxidising agents.

What is the structure of the compound formed by dehydration of **E**?

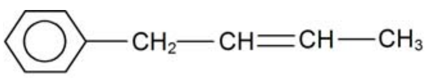
A



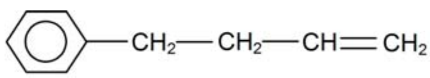
C



B



D



23 The table shows the results of simple tests on compound **G**.

Reagent	Result
2,4-dinitrophenylhydrazine	Orange precipitate
Ammoniacal silver nitrate	Silver mirror

From the results of the tests, what could **G** be?

A $CH_3CH_2COCH_3$

B $CH_3CH(OH)CH_3$

C $CH_3CH_2CH_2CHO$

D $CH_3CH_2CH_2COOH$

24 Which compound is a product of hydrolysis of $CH_3CO_2C_3H_7$ by boiling hydrochloric acid?

A CH_3COCl

B $C_3H_7CO_2H$

C C_3H_7COCl

D C_3H_7OH

25 A student attempted to perform a reaction on 3 different organic compounds using different reducing agents. He recorded his results as shown in the table below:

Test	Compounds	H_2/Ni	$LiAlH_4$	$NaBH_4$
I	$CH_3CH_2COCH_3$	√	√	√
II	$CH_3CH{=}CHCH_3$	√	**X**	**X**
III	CH_3CH_2COOH	**X**	√	**X**

Which of the test results are correct?

A I & II

B I & III

C II & III

D I, II & III

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and **3** are correct	**1** and **2** are correct	**2** and **3** are correct	**1** only is correct.

26 Which statement(s) about a 12.0 g sample of ^{12}C are correct?

1 The number of atoms in the sample is 6.02 x 10^{23}.

2 The number of atoms in the sample is the same as the number of atoms in 4.0 g of helium.

3 The number of atoms in the sample is the same as the number of atoms in 28.0 g of nitrogen gas.

(B)

27 Which of the following option(s) **must** be true of a weak acid which is relatively strong?

1 It has a high K_a value.

2 It is an acid with high concentration.

3 Its conjugate base is strong.

(D)

28 Which of the following trend(s) concerning Period 3 elements from Na to C*l* are **true**?

1 There is a change from metallic behaviour to non-metallic behaviour.

2 Their compounds show an increase in the maximum oxidation number across the period.

3 The melting points of the elements decrease across the period.

(B)

29 In an experiment, the complete combustion of a hydrocarbon yields 3.52 g of carbon dioxide and 1.44 g of water. Assume that all measurements were obtained at same temperature and pressure.

Which of the following molecular formula(e) would fit these data?

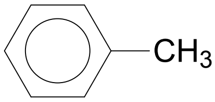
1 C_2H_4

2 C_4H_8

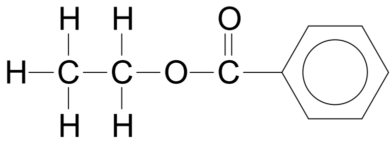
3 C_6H_6

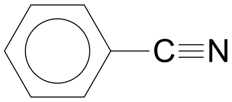
(B)

30 Which of the following compound(s) will form potassium benzoate, $C_6H_5COO^{-}K^{+}$ when refluxed in the presence of aqueous potassium hydroxide?

1 

and $KMnO_4$

2 

3 

(A)

1	**C**	11	**B**	21	**B**
2	**C**	12	**A**	22	**A**
3	**A**	13	**A**	23	**C**
4	**D**	14	**A**	24	**D**
5	**C**	15	**B**	25	**D**
6	**C**	16	**B**	26	**B**
7	**A**	17	**D**	27	**D**
8	**C**	18	**B**	28	**B**
9	**A**	19	**C**	29	**B**
10	**D**	20	**D**	30	**A**
A: 8 B: 8 C: 7 D: 7					